

ID	TYPE	Description	Priority	ACCEPTANCE CRITERIA	Rationale
FR-001	Monitoring	The system shall ping micro-service health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	High	Pass: Downtime recorded and alert dispatched on 3 consecutive failed health probes. Fail: Unreachable service marked healthy.	Real-time health visibility is the core value proposition of the portal; stale status data would let outages go unnoticed.
FR-002	Catalog Management	The system shall allow a DevOps Engineer or system architect to register a new micro-service with meta-data (name, owner, end point URL, repository link, upstream/downstream dependencies).	High	Pass: service appears in the catalogue immediately after submission with all fields persistent. Fail: service saved with missing/invalid and point URL.	The catalogue count map dependencies or run health checks, and services It doesn't know about – registration is the entry point for every other feature.
FR- 003	Visualization	The System shall render an interactive dependency graph showing all registered services as nodes and the dependency relationships as edges.	Medium	Pass: Selecting a node highlights, its direct upstream/downstream dependencies. Fail: graph fails to load or omits registered services.	System architects need to see blasted is before making changes to a shared service.
FR- 004	Documentation	The system shall aggregate and display each service's, API documentation (e.g., OpenAPI/ Swagger), searchable by service name or tag	Medium	Pass: Searching a service name returns its current API doc within 2 seconds. Fail: Search returns stale or missing documentation.	Centralized docs reduce the time engineer spend hunting across repos for API contracts.
FR -005	Alerting	System shall dispatch a downtime alert, notification (email/ Slack) to subscribed DevOps engineers when a service reaches 3 consecutive failed health probes	High	Pass: I got received within 60 seconds of the third consecutive failure, containing service name and last-healthy timestamp. Fail: no alert sent, or alert sent on a single transient failure.	Alerting is what turns passive monitoring into actionable incident response.

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NFR-001	Performance and security	The service catalogue dependency graph you must render interactive node architecture is containing up 200 services smoothly.	High	Pass: Benchmarking tests confirm target latency (<2s render) and security standards (authenticated access only) under simulated peak load.	Large organizations can have hundreds of micro services; a lag or insecure graph viewer defeats the tools purpose.
NFR-002	Reliability & Availability	The health check pinger service itself must maintain 99.9% uptime and, after a restart, resume monitoring within 60 seconds without losing previously computing 24 hour data	High	Past: simulated service restart recovers monitoring within 60 seconds with availability history intact. Fail: restart causes availability, data loss or monitoring gap greater than 60 seconds.	A monitoring system that itself goes down silently, defeats its own purpose ("who monitors the monitor").